MID TERM EXAMINATION

Subject: computer networks

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 28 | Date: 14-03-2026

Total Questions: 10 | *Answer all questions.*

Name: ____________________________ **Roll No:** _________________________

Student Sign: ___________________ **Teacher Sign:** ___________________

SECTION A: MCQS

1. Which of the following best describes an Abstract Data Type (ADT)? **[2 Marks]**

- a) A data structure that stores primitive data types only.
- b) A concrete implementation of a data structure.
- c) A mathematical model that specifies the data stored and the operations that can be performed on that data, without specifyi
- d) A data type that can only be accessed sequentially.

2. What is the primary disadvantage of using an array-based implementation for a stack or queue compared to a linked list implementation? **[2 Marks]**

- a) Slower access time for elements.
- b) Difficulty in implementing basic operations like push/pop or enqueue/dequeue.
- c) Fixed size limitation, leading to potential overflow or underflow issues if not dynamically resized.
- d) Higher memory consumption due to pointers.

3. In the context of binary trees, what is the 'height' of a tree? **[2 Marks]**

- a) The total number of nodes in the tree.
- b) The length of the longest path from the root node to any leaf node.
- c) The maximum number of children any node has.
- d) The number of levels in the tree excluding the root level.

4. Which of the following is NOT a linear data structure? **[2 Marks]**

- a) Linked List
- b) Stack
- c) Queue
- d) Binary Tree

5. Consider an algorithm with a time complexity of O(n^2). If the input size 'n' doubles, how does the execution time approximately change? **[3 Marks]**

- a) It doubles.
- b) It quadruples.
- c) It remains the same.
- d) It increases logarithmically.

6. In a singly linked list, which of the following operations has a time complexity of O(1) in the worst case, assuming a pointer to the head node is available? **[3 Marks]**

a) Accessing the k-th element.
b) Searching for a specific element.
c) Deleting the last element.
d) Inserting a new node at the beginning of the list.

7. Which of the following data structures is most suitable for implementing a "undo" functionality in a text editor? **[3 Marks]**

a) Queue
b) Array
c) Stack
d) Hash Table

8. What is the worst-case time complexity for inserting an element into a Binary Search Tree (BST) of height 'h'? **[3 Marks]**

a) O(1)
b) O(log h)
c) O(h)
d) O(h log h)

9. A circular queue with a maximum capacity of N elements is implemented using an array. If front points to the first element and rear points to the last element, which condition indicates that the queue is full? **[4 Marks]**

a) `front == rear`
b) `front == (rear + 1) % N`
c) `rear == front - 1`
d) `front == 0 && rear == N - 1`

10. Given the following inorder and preorder traversals of a binary tree, what would be its postorder traversal?

Inorder: D B E A F C

Preorder: A B D E C F **[4 Marks]**

a) D E B F C A
b) D B E A F C
c) A B D E C F
d) A D B E F C

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